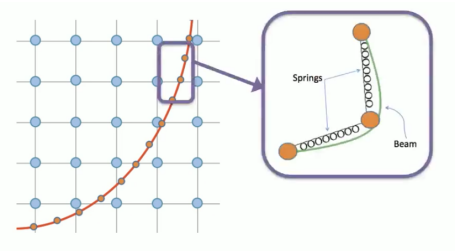
MOTIVATION FOR THE IMMERSED BOUNDARY METHOD

Classical Peskin formulation, tradeoffs, and method selection

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Fluid Dynamics and IBM Theory • Module 1



Schematic after the Wikimedia Commons figure *ImmersedBoundary grid* (Immersed boundary method article).

Claude (Anthropic) was used to assist with formatting, organization, and clarification of the instructor's original notes.

WHAT IS THE IMMERSED BOUNDARY METHOD?

The immersed boundary (IB) method is a computational framework for fluid–structure interaction in which

- fluid momentum is described on a fixed Eulerian grid;
- structure position and mechanics are described with moving Lagrangian coordinates;
- integral transforms transfer force to the fluid and velocity back to the structure.

$$\text{structure force} \xrightarrow{S} \text{fluid force},$$

$$\text{fluid velocity} \xrightarrow{J} \text{structure motion}.$$

The spreading operator S and the interpolation operator J are defined below: continuum forms on the “Continuum force and velocity coupling” slide, discrete forms S_h, J_h on the slide after it.

Introduced by Charles S. Peskin for cardiac fluid dynamics, the framework now supports many thin-boundary and volumetric FSI models.

WHY USE AN IMMERSED FORMULATION?

Body-fitted fluid meshes

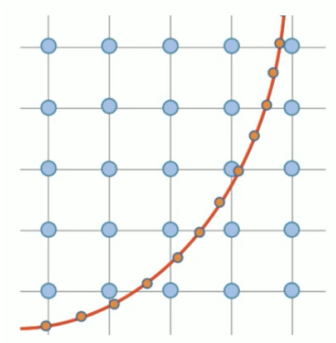
- conform to the moving fluid–structure interface;
- can resolve interfacial flow and stress sharply;
- may require mesh smoothing or remeshing when motion, deformation, or contact degrades mesh quality.

The immersed alternative

- keeps the Eulerian fluid grid fixed;
- moves a nonconforming Lagrangian representation of the structure through that grid;
- couples the two descriptions by transferring force and velocity.

> Geometric flexibility comes from the nonconforming fluid and structure descriptions. The Lagrangian structure is still tracked explicitly.

THE CLASSICAL PESKIN IB METHOD



Blue: fixed Eulerian grid. Orange: moving Lagrangian points. Schematic after the Wikimedia Commons figure *ImmersedBoundary grid* (Immersed boundary method article).

- Charles S. Peskin introduced the method to study flow around heart valves in the 1970s.
- The classical formulation couples an elastic boundary to a viscous incompressible fluid.
- Velocity, pressure, and momentum use an Eulerian description. Structure position and elasticity use a Lagrangian description.
- Later IB extensions treat volumetric solids, active materials, prescribed motion, and constrained rigid bodies.

Applications include cardiac flow, cilia and flagella, animal locomotion, flexible beams, and membranes.

Peskin, *J. Comput. Phys.* 10 (1972) and 25 (1977); Peskin, *Acta Numerica* 11 (2002); Griffith and Patankar, *Annu. Rev. Fluid Mech.* 52 (2020).

REPRESENTATIVE APPLICATIONS

Biological fluid mechanics

- heart valves, chambers, and vascular flow;
- cilia- and flagella-driven transport;
- swimming by fish, jellyfish, and microorganisms;
- flexible airways and tissue–fluid interaction.

Engineering and soft matter

- flapping flags, membranes, and beams;
- flow-induced vibration and mixing;
- bio-inspired propulsion and soft robotics;
- compliant or porous surfaces, with extensions tailored to the required physics.

> The common feature is not a single Reynolds-number regime; it is a moving or deforming interface that benefits from a nonconforming fluid grid.

WHAT THE CLASSICAL IB METHOD IS—AND IS NOT

It is

- a force-based FSI formulation;
- an Eulerian–Lagrangian coupling framework;
- a method whose standard discretization uses a regularized delta kernel;
- compatible with large structural motion.

It is not

- a body-fitted or intrinsically sharp-interface method;
- a method that discards elasticity or stress;
- automatically topology changing—the structural representation must model any change;
- restricted to one constitutive law.

> At the continuum level no slip is pointwise. “Weak enforcement” describes the regularized discrete interaction, not the singular continuum equations.

CONTINUUM FORCE AND VELOCITY COUPLING

Let $q \in \mathcal{U}$ label a material point and let $\boldsymbol{X}(q,t)$ be its position in the fluid domain Ω.

Spread structural force to the fluid

$$\boldsymbol{f}(\boldsymbol{x},t) = \int_{\mathcal{U}} \boldsymbol{F}(q,t)\,\delta(\boldsymbol{x} - \boldsymbol{X}(q,t))\; dq.$$

$\boldsymbol{F}$ is a force density per unit material measure. The Dirac distribution places that force at the current structure location.

Interpolate fluid velocity to the structure

$$\boldsymbol{U}(q,t) = \frac{\partial \boldsymbol{X}}{\partial t} = \int_{\Omega} \boldsymbol{u}(\boldsymbol{x},t)\,\delta(\boldsymbol{x} - \boldsymbol{X}(q,t))\; d\boldsymbol{x}.$$

For continuous $\boldsymbol{u}$, the Dirac identity gives

$$\frac{\partial \boldsymbol{X}}{\partial t}(q,t) = \boldsymbol{u}(\boldsymbol{X}(q,t),t)\,.$$

> At the continuum level, the kinematic equation imposes pointwise no slip on an impermeable material interface. It is not merely an averaged boundary condition.

In operator form, $\boldsymbol{f} = S[\boldsymbol{X}]\,\boldsymbol{F}$ and $\partial_t \boldsymbol{X} = J[\boldsymbol{X}]\,\boldsymbol{u}$ with $J = S^*$, the L^2 adjoint of S. Adapted from Griffith and Patankar (2020), §2.1.1, Eqs. 1–4, and Peskin (2002).

THE STANDARD DISCRETE IB COUPLING

The numerical method replaces δ by a regularized kernel δ_h whose support is a few Eulerian grid cells wide:

$$\boldsymbol{f}_h = S_h \boldsymbol{F}_h, \qquad \boldsymbol{U}_h = J_h \boldsymbol{u}_h.$$

- S_h spreads Lagrangian forces to nearby grid points.
- J_h interpolates grid velocity to Lagrangian interaction points.
- Kernel support and moment conditions control smoothness, localization, and grid dependence.

Choose J_h as the weighted discrete adjoint of S_h so that the transferred power is consistent:

$$\langle \boldsymbol{u}_h, S_h \boldsymbol{F}_h \rangle_E = \langle J_h \boldsymbol{u}_h, \boldsymbol{F}_h \rangle_L .$$

Regularization spreads a singular force layer and associated jump fields over $O(h)$. The Lagrangian boundary itself remains explicitly represented.

WHY ADJOINT OPERATORS TRANSFER POWER CONSISTENTLY

With Eulerian grid spacing h and Lagrangian quadrature weights Δq_l, define

$$\langle \boldsymbol{u}_h, \boldsymbol{f}_h \rangle_E = \sum_{\boldsymbol{x}} \boldsymbol{u}_h(\boldsymbol{x}) \cdot \boldsymbol{f}_h(\boldsymbol{x})\, h^3, \qquad \langle \boldsymbol{U}_h, \boldsymbol{F}_h \rangle_L = \sum_{l} \boldsymbol{U}_l \cdot \boldsymbol{F}_l\, \Delta q_l.$$

- Power the spread force does on the fluid: $P_E = \langle \boldsymbol{u}_h, S_h \boldsymbol{F}_h \rangle_E$.
- Power the structural force does at the interpolated structure velocity: $P_L = \langle J_h \boldsymbol{u}_h, \boldsymbol{F}_h \rangle_L$.
- If $J_h = S_h^*$, then $P_E = P_L$ for every $\boldsymbol{u}_h$ and $\boldsymbol{F}_h$: the transfer step neither creates nor destroys energy.

Here $\langle \cdot, \cdot \rangle_E$ and $\langle \cdot, \cdot \rangle_L$ are the discrete Eulerian and Lagrangian inner products induced by the quadrature rules. See Griffith and Patankar (2020), §3.3, who write these operators as S_{IB} and J_{IB} with $J_{\mathrm{IB}} = S_{\mathrm{IB}}^* = h^3 M^{-1} S_{\mathrm{IB}}^{\mathsf{T}}$, where M is the diagonal matrix of Lagrangian quadrature weights (for uniform weights Δq, $J_{\mathrm{IB}} = (h^3/\Delta q)\, S_{\mathrm{IB}}^{\mathsf{T}}$).

WHAT REGULARIZED IB COUPLING BUYS AND COSTS

Strengths

- No body-conforming fluid mesh or fluid-grid remeshing
- Large structural motion and deformation
- Thin immersed boundaries and transient contact
- Efficient Cartesian-grid solvers and adaptive mesh refinement
- Fiber, surface, and volumetric structural models

Numerical challenges

- Pressure and traction jumps are smoothed at finite h
- Interfacial forces and stresses may converge slowly
- Coarse interaction-point spacing can cause leakage or volume error
- Stiff elastic forces can severely restrict explicit time steps

AMR targets spatial resolution. Implicit or semi-implicit coupling targets stiffness. Sharp-interface corrections target jump accuracy. Finite-element structure models broaden constitutive and geometric choices.

ACCURACY AND REGULARIZED BOUNDARY COUPLING

At finite grid spacing, δ_h averages information across a stencil several cells wide.

- The interpolated velocity need not equal the pointwise trace of a sharply resolved field.
- Pressure and stress jumps are represented over an $O(h)$ layer.
- Kernel moments, support, and smoothness influence conservation and grid dependence.

> Accuracy depends jointly on
>
> h, Δs or quadrature density, Δt,
>
> the fluid solver, the structural discretization, and solution regularity.

A refinement study must vary every resolution that controls the coupled approximation.

COMMON FAILURE MODES AND PRACTICAL CHECKS

Interaction resolution

- Interaction points too far apart can leak.
- A very fine structural mesh may add stiffness or poor conditioning without improving the fluid resolution.
- Different spreading and interpolation kernels break the usual adjoint power identity.

Time integration and verification

- Explicit elastic forces can impose severe time-step limits.
- Monitor volume, deformation, net force, and transferred power separately.
- Refine Eulerian, Lagrangian, quadrature, and temporal resolutions systematically.

> Many apparent “IB failures” are mismatches among the grid, interaction points, kernel, and time integrator rather than failures of the continuum formulation.

With nodal quadrature, a structure leaks when its Lagrangian nodes become too far apart relative to h (Peskin 2002). Non-nodal (e.g. Gaussian) quadrature lets the structural mesh be coarse provided the net of quadrature points is dense enough (Griffith and Luo 2017; Griffith and Patankar 2020, §3.3).

WHEN IS CLASSICAL IB A GOOD CHOICE?

Strong fit

- Large or complicated structural motion
- Thin, flexible, or actively deforming structures
- Transient contact without a body-fitted fluid mesh
- Quantities of interest that tolerate regularized interfacial fields

Requires special care

- Pointwise tractions or high-order interface stresses
- Thin high-Reynolds-number boundary layers
- Nearly rigid constraints or severe stiffness
- Tearing, merging, or reconnection, which require an explicit topology model

> High Reynolds number is not a formal exclusion. Published immersed FSI examples reach $\mathrm{Re} \approx 2 \times 10^4$, but thin shear layers and turbulence still require adequate resolution and modeling.

Griffith and Patankar, *Annu. Rev. Fluid Mech.* 52 (2020); the $\mathrm{Re} \approx 2 \times 10^4$ example is the bioprosthetic heart-valve application of their §5.3.1.

DISCRETIZATION AND CONVERGENCE CHECKLIST

Interaction geometry

- Resolve the structure with enough Lagrangian interaction points or quadrature points relative to h.
- Coarse nodal coupling can leak. Excessively fine structural meshes add cost and may worsen stiffness or conditioning.
- Preserve weighted adjointness between spreading and interpolation.

Verification

- Refine h, the Lagrangian discretization or quadrature, and Δt consistently.
- Check velocity, deformation, volume, loads, and energy transfer separately.
- Use an implicit or semi-implicit treatment when elastic stiffness controls stability.

> If the support of δ_h scales with h and the coupled scheme is stable and consistent, refinement recovers the singular IB formulation. Convergence rates depend on solution regularity; standard regularized methods are often only first order near stress jumps.

Peskin (2002); Griffith and Patankar (2020); Griffith and Luo (2017).

BODY-FITTED FSI: WHY CONFORMING MESHES ARE ATTRACTIVE

A body-fitted fluid mesh conforms to the moving interface.

- Boundary conditions and traction continuity can be imposed on the represented surface.
- Boundary layers and interfacial stresses can be resolved without regularized smearing.
- ALE, moving-mesh finite element, and moving-mesh finite volume schemes are common examples.

Geometric bargain

sharp, conforming interface

fluid-mesh motion and quality control

BODY-FITTED FSI: COST UNDER LARGE MOTION

Typical numerical work

- solve for mesh motion or smoothing;
- detect distortion and remesh when needed;
- transfer fields between successive meshes;
- maintain coupled fluid–solid interface data structures.

Best fit

- interface accuracy is central;
- motion remains compatible with mesh quality;
- the added geometric machinery is justified by the quantities of interest.

> Large deformation is possible in sophisticated body-fitted schemes; it is the mesh-management burden, not a universal deformation cutoff, that distinguishes the tradeoff.

COMPARISON WITH OTHER FSI APPROACHES

Method family	Fluid and interface treatment	Typical numerical challenge
Regularized IB	Fixed, nonconforming grid; δ_h spreading and interpolation	Smeared jump fields, interfacial accuracy, leakage, and stiffness
ALE / body-fitted	Moving, boundary-conforming fluid mesh	Mesh quality, mesh motion, and possible remeshing under large motion
Immersed interface / sharp immersed	Fixed grid; interface-aware jump corrections	Geometric reconstruction and jump-dependent stencils or solvers
Fictitious domain / penalization	Fixed background grid; distributed constraints or penalties	Conditioning, penalty or modeling error, and method-specific coupling

> The deformation a method can handle depends on its interface representation, mesh strategy, solver, and implementation. Small, moderate, and large are not reliable method-family labels.

THE IMMERSED INTERFACE METHOD

The immersed interface method (IIM) keeps a fixed Eulerian grid and incorporates analytically derived jump conditions into nearby stencils.

- It avoids regularized smearing of jump fields.
- It can improve near-interface velocity, pressure, deformation, and traction accuracy.
- It adds geometry reconstruction and jump-dependent stencils or solvers.

Define $[\![a]\!] = a^+ - a^-$ and take $\boldsymbol{n}$ from Ω^- to Ω^+. If $\boldsymbol{f}_\Gamma$ is the interface-on-fluid force in $+\boldsymbol{f}_\Gamma \delta_\Gamma$, then

$$[\![\boldsymbol{u}]\!] = 0, \qquad [\![\boldsymbol{\sigma}]\!]\, \boldsymbol{n} = -\boldsymbol{f}_\Gamma.$$

Traction balance determines jumps in pressure and velocity derivatives. Pressure may be continuous or discontinuous, depending on the interfacial force.

> Sharp means resolving jumps without regularized smearing; it does not mean the computed solution is exact. Large or topology-changing motion still depends on the interface and geometry algorithms.

Li and Lai, *J. Comput. Phys.* 171 (2001); Griffith and Patankar (2020).

WORKED STEP: THE SIGN OF THE STRESS JUMP

Consider a steady one-dimensional balance with a point force at an interface $x = 0$, normal $\boldsymbol{n} = +\boldsymbol{e}_x$ pointing from $\Omega^- = \{x < 0\}$ to $\Omega^+ = \{x > 0\}$:

$$0 = \frac{\partial \sigma}{\partial x} + f_\Gamma \, \delta(x).$$

Integrate across $(-\epsilon, \epsilon)$ and let $\epsilon \to 0$:

$$\sigma(0^+) - \sigma(0^-) + f_\Gamma = 0 \qquad \Longrightarrow \qquad \boxed{[\![\sigma]\!] = -f_\Gamma}.$$

In several dimensions the same pillbox argument gives $[\![\boldsymbol{\sigma}]\!]\,\boldsymbol{n} = -\boldsymbol{f}_\Gamma$. With $\boldsymbol{\sigma} = -p\boldsymbol{I} + \mu(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^{\mathsf{T}})$, the normal component yields the pressure jump and the tangential components yield jumps in velocity derivatives.

> If the force pushes the fluid toward Ω^+ ($f_\Gamma > 0$), the stress on the $+$ side is lower; for a fluid at rest ($\sigma = -p$) this means $p(0^+) = p(0^-) + f_\Gamma$, so the force compresses the fluid ahead of it.

Pillbox derivation in the spirit of Peskin (2002) and Li and Lai (2001).

IMMERSED INTERFACE METHODS: STRENGTHS AND TRADEOFFS

Strengths

- fixed background grid;
- sharp representation of pressure and stress jumps;
- potentially higher near-interface accuracy;
- no regularized force layer in the target continuum description.

Added requirements

- interface geometry and normals;
- jump conditions consistent with the physical model;
- corrected stencils or specialized solvers;
- robust handling of large motion and topology changes.

> IIM prioritizes sharp jump accuracy; regularized IB prioritizes a simple, robust transfer mechanism. The right choice follows the observable and geometry.

OPEN-SOURCE SOFTWARE AND TUTORIALS

IBAMR

- Distributed-memory C++ framework
- Cartesian adaptive mesh refinement
- Examples for classical IB, IBFE, constrained IB, and IIM in two and three dimensions

`ibamr.github.io`
Griffith et al., *J. Comput. Phys.* 223 (2007).

IB2d

- MATLAB and Python implementations in two dimensions
- More than 75 examples for teaching and research
- Introductory video tutorials linked from the repository

`github.com/nickabattista/IB2d`
Battista, Strickland, and Miller, *Bioinspir. Biomim.* 12 (2017).

> Nick Battista's introduction to the IB method is the first tutorial linked by IB2d. Course-specific lectures by Matea Santiago may require Google Drive access.

WHY THE OPEN-SOURCE ECOSYSTEM MATTERS

For learning and verification

- inspect the actual spreading and interpolation operators;
- reproduce benchmark problems;
- compare kernels, solvers, and time integrators;
- trace continuum notation into implementation.

For research

- prototype new constitutive or active-force models;
- use tested parallel solvers and AMR infrastructure;
- document algorithmic choices for reproducibility;
- separate model development from low-level solver development.

> Software availability is a practical advantage, not a mathematical accuracy claim. Validation is still problem- and implementation-specific.

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> Free copies of the second and third papers are on PubMed Central (PMC7531444 and PMC5650596). The repository's `papers/README.md` lists links for all of them.

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- Wikimedia Commons, figure “ImmersedBoundary grid,” used in the Wikipedia article “Immersed boundary method,” https://en.wikipedia.org/wiki/Immersed_boundary_method. Source of the schematics `ImmersedBoundary_grid.png` and `ImmersedBoundary_grid2.png`; licence to be confirmed.